Auteur: Michael Livchits
Studierichting: Informatica
Oefeningenreeks 6E nr. 11.3

$f(x) = x^8 + 8x^7 + 12x^6 - 40x^5 - 74x^4 + 120x^3 + 108x^2 - 216x + 81$

$f'(x) = 8x^7 + 56x^6 + 72x^5 - 200x^4 - 296x^3 + 360x^2 + 216x - 216$

Vind de nulpunten:

	8	56	72	−200	−296	360	216	−216
1		8	64	136	−64	−360	0	216
	8	64	136	−64	−360	0	216	0
−1		−8	−56	−80	144	216	−216	
	8	56	80	−144	−216	216	0	
1		8	64	144	0	−216		
	8	64	144	0	−216	0		
1		8	72	216	216			
	8	72	216	216	0			
−3		−24	−144	−216				
	8	48	72	0				
−3		−24	−72					
	8	24	0					

$8x + 24 = 0$

$x = -3$

$\Rightarrow f'(x) = 8(x-1)^3(x+3)^3(x+1)$

$f''(x) = 56x^6 + 336x^5 + 360x^4 - 800x^3 - 888x^2 + 720x + 216$

$f''(1) = 0$, $f''(-3) = 0$, $f''(-1) = 56 - 336 + 360 + 800 - 888 - 720 + 216 = -512$

$f''(-1) < 0 \Rightarrow$ lokale max.

$f'''(x) = 336x^5 + 1680x^4 + 1440x^3 - 2400x^2 - 1776x + 720$

$f'''(1) = 0$, $f'''(-3) = 0$

$f^{iv}(x) = 1680x^4 + 6720x^3 + 4320x^2 - 4800x - 1776$

$f^{iv}(1) = 1680 + 6720 + 4320 - 4800 - 1776 = 6144$

$f^{iv}(1) > 0 \Rightarrow$ lokale min.

$f^{iv}(-3) = 136080 - 181440 + 38880 + 14400 - 1776 = 6144$

$f^{iv}(-3) > 0 \Rightarrow$ lokale min.

Dus de punten zijn: $f(-1) = 256,\ f(1) = 0,\ f(-3) = 0$

$\Rightarrow$ lokaal maximum: $(-1, 256)$

$\Rightarrow$ lokale minima: $(1, 0)$ en $(-3, 0)$

References